

# Cognitive Pathologies: Confounding Factors

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## Abstract

Accurately interpreting cognitive test data requires addressing several challenges, including: (I) short-term variability in participant performance, (II) biases in data samples, (III) variations in participants' prior experience with cognitive testing, and (IV) spurious correlations between cognitive performance and various personal characteristics. This report outlines several methods for mitigating these issues.

The structure of the report is as follows: Section 1 presents the computational model used for data processing. Section 2 outlines a method for identifying markers indicative of current and future cognitive states of participants. Section 3 describes an approach for assessing the cognitive health of individual participants. Section 4 discusses potential avenues for further research.

## 1 Computational model

### 1.1 Algorithm selection

Three standard techniques are commonly used for making predictions based on tabular data: neural networks, regression analysis, and decision tree algorithms. While neural networks typically deliver the best performance on large datasets, they tend to overfit when applied to the MHAS data, where the ratio of data points to dependent variables is less than 30. Ridge regression effectively addresses the overfitting issue but provides suboptimal results due to the significant non-linear interactions between the dependent and target variables. Among the three decision tree algorithms—XGBoost, CatBoost, and LightGBM—testing has shown that LightGBM offers the highest accuracy.

### 1.2 Blending and cross-validation

To reduce overfitting, the data was randomly divided multiple times into training and testing subsets comprising respectively 80 and 20% of all data. Each time the training data was used to generate a new decision tree model whose predictions were blended together. The comparison between the resulting predictions and actual values in the testing subsets was used to evaluate the overall accuracy of the model.

### 1.3 Data preprocessing

Most numerical features (integers and floats) were directly input into the model. An exception was made for the education level variables, *edu\_gru\_03* and *edu\_gru\_12*. Since no participant reported a change in their education level between 2003 and 2012,

these two variables were merged into a single feature. If either *edu\_gru\_03* and *edu\_gru\_12* was missing, the other variable was used to fill in the missing value for the new variable.

For categorical variables, most were converted into integers based on the natural order of their values (e.g., “Excellent” = 1, “Good” = 2, “Fair” = 3, etc.). For two variables, *rjlocc\_m\_12* (category of longest occupation) and *j11\_12* (floor material of residence), which lacked a clear order, the categories were ranked based on the mean education level within each category.

## 1.4 Addressing sample bias

The MHAS study’s participants were initially selected to create a representative population sample. However, we found a significant sampling bias in the later stages of the research. Analysis of the 2016 data revealed that older participants, those with fewer years of education, and individuals who performed below expectation for their age and education level were substantially less likely to participate in subsequent testing.

The following technique was found to be effective in mitigating the effect of this bias on the model’s accuracy:

- Add two new binary features, *y16* and *y21*, to indicate whether the individual participated in the 2016 and 2021 tests, respectively.
- For participants who took both the 2016 and 2021 tests, calculate the difference between their test scores.
- Develop a decision tree model to predict the difference between the two scores. Use this model to predict score differences for all users, including those who participated in only one test. To prevent data leakage, make predictions for each user using only the models trained on subsets of data that exclude the user being predicted.
- For participants who have 2016 test scores, estimate their 2021 scores by adding the predicted score difference to their 2016 scores. Combine these estimates with actual 2021 scores to train the final model for predicting test scores in 2021.
- For participants who have 2021 test scores, estimate their 2016 scores by subtracting the predicted score difference from their 2021 scores. Combine these estimates with actual 2016 scores to train the final model for predicting test scores in 2016.

## 2 Determinants of health

Cognitive test scores in elderly individuals are primarily influenced by two key factors: age and presenescence cognitive performance (i.e., peak performance before the onset of cognitive decline). These factors strongly correlate with various medical conditions, mental activities, and social characteristics, resulting in significant correlations between nearly all features in our dataset and cognitive test scores. Some of these relationships, such as the positive correlation between tobacco and alcohol use and cognitive scores, appear counterintuitive.

To differentiate features with a potential causal link to cognitive decline from spu-

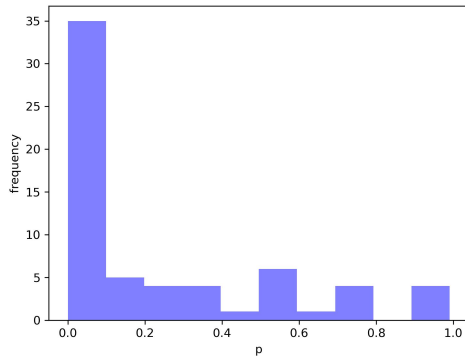


Figure 1: Distribution of p-values for 64 binary features (from 2012) when test scores are controlled for age and education levels.

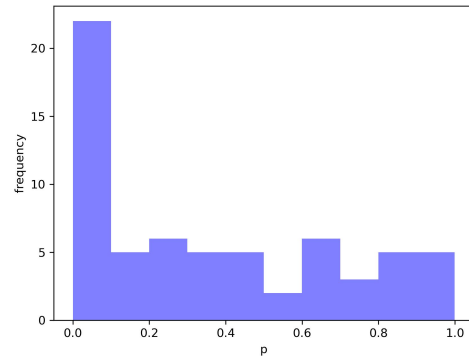


Figure 2: Distribution of p-values for 64 binary features (from 2012) when test score changes (from 2016 to 2021) are controlled for age and education levels.

rious correlations, it is necessary to control for age and presenescent cognitive performance. However, as neither was directly included in our dataset, binned age group and binned education level were chosen as proxies. Since the binned age group is only a coarse approximation of age, controlling for it may still preserve spurious correlations between cognitive scores and features strongly correlated with age. To address this, an additional variable was constructed to predict participants' age groups based on all features excluding the variables for binned age group themselves. Similarly, an additional variable was constructed to predict participants' education levels based on all other variables.

After controlling for binned age, education, and their predicted values, approximately half of the features, including tobacco and alcohol use, no longer exhibited statistically significant correlations with cognitive scores<sup>1</sup>. Similarly, only about one-third of the features showed significant correlations with changes in cognitive scores between 2016 and 2021 (see Figures 1 and 2 and Tables 1 and 2 for p-curves and regression coefficients).

<sup>1</sup> Unlike most developed countries, tobacco and alcohol use in Mexico are positively correlated with education and negatively correlated with age (e.g., Buitenhuis et al. 2010), which explains the positive correlation with the cognitive test scores.

|                   | p-value               | Coef.  |
|-------------------|-----------------------|--------|
| adl_walk_12       | $5.7 \times 10^{-4}$  | -10.47 |
| adl_bath_12       | $1.3 \times 10^{-3}$  | -14.98 |
| adl_eat_12        | $6.4 \times 10^{-3}$  | -13.31 |
| adl_toilet_12     | $8.3 \times 10^{-3}$  | -7.81  |
| iadl_money_12     | $1.0 \times 10^{-5}$  | -21.38 |
| iadl_meds_12      | $4.2 \times 10^{-4}$  | -19.03 |
| iadl_shop_12      | $9.2 \times 10^{-4}$  | -8.15  |
| iadl_meals_12     | $6.2 \times 10^{-6}$  | -18.57 |
| hard_12           | $2.2 \times 10^{-4}$  | -4.95  |
| cesd_depressed_12 | $4.5 \times 10^{-3}$  | -3.89  |
| diabetes_12       | $4.9 \times 10^{-5}$  | -6.09  |
| stroke_12         | $8.8 \times 10^{-4}$  | -14.78 |
| test_pres_12      | $2.3 \times 10^{-3}$  | 4.76   |
| imss_12           | $2.1 \times 10^{-6}$  | 6.01   |
| care_child_12     | $1.5 \times 10^{-6}$  | 6.79   |
| volunteer_12      | $1.1 \times 10^{-4}$  | 8.39   |
| games_12          | $1.8 \times 10^{-5}$  | 7.38   |
| table_games_12    | $7.5 \times 10^{-5}$  | 7.03   |
| comms_tel_comp_12 | $8.3 \times 10^{-6}$  | 6.18   |
| tv_12             | $1.6 \times 10^{-6}$  | 11.13  |
| sewing_12         | $1.8 \times 10^{-11}$ | 9.29   |
| n_living_child_12 | $7.8 \times 10^{-5}$  | 2.13   |
| n_adl_12          | $6.8 \times 10^{-4}$  | -3.38  |
| n_iadl_12         | $3.1 \times 10^{-8}$  | -7.61  |
| bmi_12            | $2.5 \times 10^{-5}$  | 3.41   |
| decis_famil_12    | $8.4 \times 10^{-3}$  | -3.22  |
| decis_personal_12 | $2.1 \times 10^{-4}$  | -6.25  |
| rrfcntx_m_12      | $7.8 \times 10^{-6}$  | -0.89  |
| rsocact_m_12      | $3.9 \times 10^{-7}$  | -0.92  |

Table 1: Relationship between the MHAS survey variables and test scores (only  $p < 0.01$  values are included).

|                  | p-value              | Coef.  |
|------------------|----------------------|--------|
| migration_12     | $2.2 \times 10^{-3}$ | -7.32  |
| depressed_12     | $7.9 \times 10^{-3}$ | -3.69  |
| sad_12           | $2.5 \times 10^{-3}$ | -4.07  |
| diabetes_12      | $1.6 \times 10^{-3}$ | -4.94  |
| arthritis_12     | $9.8 \times 10^{-5}$ | -7.01  |
| test_chol_12     | $3.3 \times 10^{-3}$ | -4.20  |
| insur_private_12 | $9.5 \times 10^{-3}$ | -11.44 |
| insured_12       | $1.2 \times 10^{-3}$ | -6.38  |
| sgender_12       | $9.6 \times 10^{-5}$ | 5.98   |
| n_illnesses_12   | $4.7 \times 10^{-5}$ | -2.87  |

Table 2: Relationship between the MHAS survey variables and changes in test scores between 2016 and 2021 (only  $p < 0.01$  values are included).

Most feature correlations with test scores followed expected patterns, with medical insurance being a notable exception. While having medical insurance (public or private) generally correlates positively with cognitive scores, this relationship inverted after adjusting for age and education. Notably, private medical insurance emerged as the strongest predictor of cognitive decline between 2016 and 2021. A plausible explanation is that individuals anticipating significant health deterioration are more likely to invest in medical insurance.

### 3 Evaluating cognitive health of individual patients

The MHAS data indicates that neither low test scores nor a sharp decline in test scores between the 2016 and 2021 measurements can reliably diagnose cognitive pathologies due to several confounding factors:

- **Presenescent cognitive performance:** Individuals in the early stages of Alzheimer's disease (AD) who begin their cognitive decline from a higher baseline may still outperform the average healthy person. Figure 3 highlights this issue, showing that highly educated individuals tend to achieve significantly higher scores than those without formal education, regardless of age.
- **Varying levels of familiarity with cognitive testing:** Figure 4 depicts the average change in test scores between 2016 and 2021 for participants with different education levels. As expected, test performance declines with age among individuals with longer formal education. However, for those on the lowest education level, familiarity with the test format after the initial assessment compensated for cognitive decline, resulting in higher average scores in 2021 compared to 2016.
- **Short-term variability in test results:** Figure 5 demonstrates that participants who overperformed or underperformed relative to the predictions of a decision tree model in 2016 tended to regress approximately one-third of the way toward the mean in 2021. The magnitude of this regression indicates that, for an average participant, the short-term variability in test results is several times greater than the actual cognitive decline observed over the five-year interval.

Given the first two of the above problems, the most reliable indicator of cognitive pathology may not be absolute test scores but rather performance relative to a peer group (i.e., individuals of similar age and education). To address the issue of short-term variance, it is more accurate to evaluate patients using predicted scores based on their survey data instead of actual test results. Figure 6 shows the relative cognitive decline for all survey participants that was predicted based on this approach.

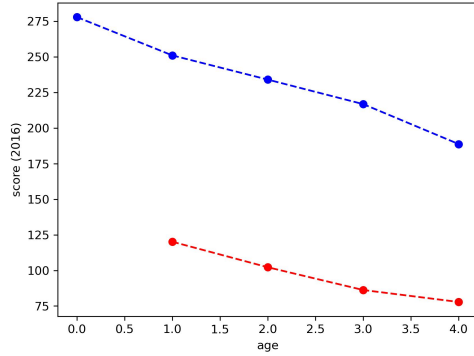


Figure 3: Average 2016 test scores for different age groups (red and blue lines correspond to participants with the lowest and the highest education levels respectively).

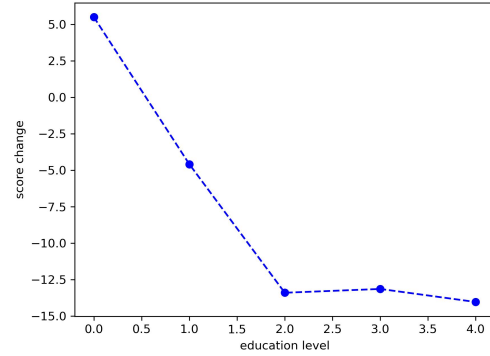


Figure 4: Average test score change from 2016 to 2021 for people with different education levels.

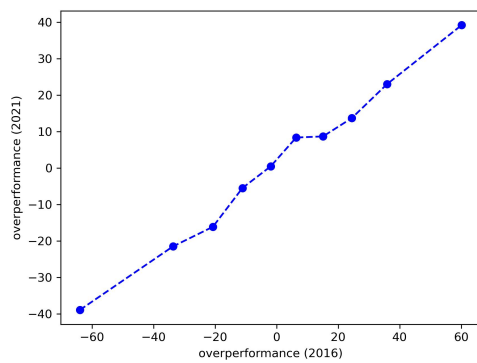


Figure 5: Test performance relative to expectations set by the survey data (the horizontal and vertical axes show participant's performance on the 2016 and 2021 tests respectively).

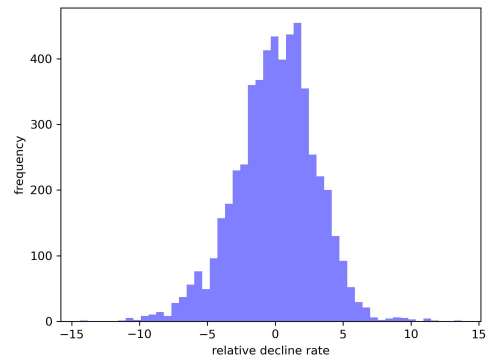


Figure 6: Expected five-year decline rate of cognitive test scores relative to participants' peer group (people with same age and education levels).

## 4 Future Directions

The accuracy of the presented models may be significantly enhanced by incorporating additional data from the MHAS survey. Specifically, instead of relying solely on the composite cognitive test score, we suggest analyzing each of its individual components separately. Since the seven components likely vary in their sensitivity to participants' education levels and cognitive health status, this approach may facilitate distinguishing differences in education from cognitive pathologies.

Based on insights gained from the MHAS data, we propose the following additions and modifications to its format:

- **Additional cognitive tests:** When participants' cognitive performance is assessed only twice, it becomes challenging to differentiate short-term variability in test results from genuine long-term changes caused by aging or progressive dementia. This limitation could be addressed by future studies that evaluate participants at least three times or by administering additional cognitive tests to current MHAS participants.
- **Additional survey questions:** The observed link between medical insurance purchases and future cognitive decline in the MHAS data suggests that participants or their families may hold valuable insights into their health that are not captured by the current survey questions. To address this, future surveys could include questions about participants' expectations regarding changes in their health and the reasons behind these expectations. Responses to such questions would provide potentially valuable information for both diagnosing individual participants and identifying new risk factors for accelerated cognitive decline.

## References

Buttenheim AM, Wong R, Goldman N, Pebley AR. *Does social status predict adult smoking and obesity? Results from the 2000 mexican national health survey*. Global Public Health. 2010;5(4):413–426.